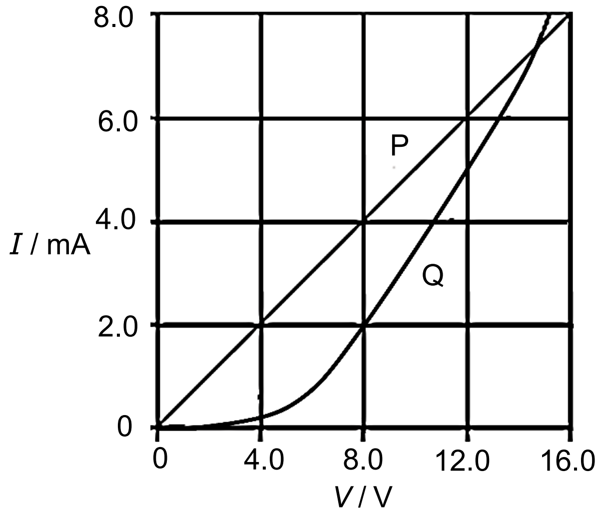
20 The I-V characteristics of two electrical components P and Q are shown below.



Which of the following statements is correct?

A P is an ohmic conductor and Q is a filament.

B Resistance of Q is always larger than resistance of P.

C At 2.0 mA, the power dissipated through Q is twice that of P.

D At the point where the two lines intersect the resistance of Q is approximately twice that of P.

Ans: C

Option A: Q is a semi-conductor diode, not filament

Option B: Resistance of Q becomes smaller than P beyond $V \approx 14$ V.

Option C: V_Q is twice V_P at 2.0 mA. Since $P = IV$, P_Q is twice P_P.

Option D: At intersection, resistance of both are equal since V and I are the same.

21 The diagram shows a network of three identical resistors connected to a battery of negligible